Covinventory ERP

Enterprise Inventory, Billing, Accounting & GST Platform — Comprehensive Feature Specification

An end-to-end technical and operational reference of all features, modules, workflows, and integrations built into the Covinventory Multi-Tenant SaaS platform.

Platform Version: 2.4.0 (Enterprise)	**Document Type:** Technical Features Specification
Architecture: Multi-Tenant Cloud ERP	**Technology Stack:** React 18, Vite, Tailwind CSS, Firestore
Target Industry: Retail, Wholesale, Distribution, Manufacturing	**Compliance:** GST (GSTR-1, GSTR-3B, E-Way Bill, E-Invoicing)
Document Status: Complete / PDF-Ready	**Last Updated:** August 2026

---

1. Executive Overview & System Architecture

Covinventory is an enterprise-grade billing, inventory, and accounting platform modeled after Indian GST and retail/wholesale operational standards. It provides business owners, counter cashiers, accountants, and warehouse supervisors with real-time operational control.

Architecture Highlights

- **Front-End Layer:** React 18 with Vite for sub-second hot reload and modular chunk compilation.
- **Design System:** Tailored Vanilla/Tailwind CSS with tabular number figures (tnum), clean borders, and dark/light mode parity.
- **Database & State:** Google Cloud Firestore NoSQL document store with multi-tenant logical partitioning by companyId.
- **Audio Synthesis:** Web Audio API 800Hz sine-wave synthesis for instant POS scan feedback without external asset dependencies.
- **Printing Engine:** Native CSS print media query formatting for 80mm thermal rolls, 58mm thermal rolls, and desktop A4/A5 invoices.

---

2. Comprehensive Feature Matrix

A consolidated functional matrix detailing Covinventory modules, application routes, operational status, and core business capabilities:

Module	Route	Status	Key Operational Capabilities
POS Fast Billing	`/pos`	Production Ready	Fullscreen kiosk terminal, hotkeys (F2, F4, F7, F9), continuous barcode scanning, split tender, thermal receipts
Tax Invoices	`/sales`, `/sales/new`	Production Ready	GST/Non-GST tax invoices, auto CGST/SGST/IGST detection, line discounts, dynamic UPI QR codes
Quotations / Estimates	`/quotations`	Production Ready	Formal price proposals with 1-click Convert to Invoice conversion pipeline
Delivery Challans	`/challans`	Production Ready	Goods dispatch transport packing slips tracking vehicle/driver logs; converts to invoice
Credit & Debit Notes	`/returns`	Production Ready	Sales Returns & Purchase Returns with automated inventory restocking and ledger balancing
Product Catalog Master	`/inventory`	Production Ready	Dual unit conversion (e.g., 1 Box = 12 pcs), HSN/SAC mapping, tax slabs, low stock alerts
Batch & Expiry (FEFO)	`/inventory`	Production Ready	Batch numbers, mfg & expiry dates, near-expiry visual warnings, FEFO picking order
Serial Number / IMEI	`/inventory`	Production Ready	Unique individual serial and IMEI tracking for electronics and consumer appliances
Stock Adjustments	`/inventory/adjustments`	Production Ready	Manual adjustment audit logs for damage, breakage, expired stock, theft, and audit variance
Multi-Godowns	`/inventory/godowns`	Production Ready	Branch warehouse locations and Inter-Godown Stock Transfer Vouchers
Barcode & QR Designer	`/inventory/barcodes`	Production Ready	Code128 and QR label designer with 12-up, 24-up, and 40-up printable sheet layouts
Purchases & Inward	`/purchases`	Production Ready	Vendor purchase bills, input tax credits, and automated stock replenishment
Cash Book & Banking	`/banking`	Production Ready	Petty cash register, multi-bank accounts, digital wallets, and Contra fund transfers

Module	Route	Status	Key Operational Capabilities
Expense Tracker	`/expenses`	Production Ready	Overhead operating expenses with GST Input Tax Credit (ITC) eligibility tracking
Payment Vouchers	`/payments`	Production Ready	Payment In (Customer receipts) & Payment Out (Vendor payouts) with multi-bill settlement
Party Ledgers (CRM)	`/customers`, `/suppliers`	Production Ready	Credit limits, payment credit terms (e.g., 30 days), interactive Statement of Accounts
GSTR-1 Reports & JSON	`/gst-reports`	Production Ready	Tables 4A, 7, 9B, 12 with direct NIC-compliant JSON and CSV portal uploads
GSTR-3B Summary	`/gst-reports`	Production Ready	Monthly consolidated Outward Tax Liability vs. Inward ITC and Net Cash Payable
NIC E-Way Bill JSON	`/gst-reports`	Production Ready	JSON generation for consignments exceeding ₹50,000 with Transporter ID and vehicle logs
E-Invoicing (IRN & QR)	`/gst-reports`	Production Ready	64-character hash generation and B2B Signed QR code simulation
Profit & Loss (P&L)	`/reports`	Production Ready	Gross Sales Revenue, Cost of Goods Sold (COGS), Operating Overheads, and Net Profit
Balance Sheet	`/reports`	Production Ready	Liquid Assets, Inventory Asset Value, Receivables, Payables, and Owner Capital Equity
Bill-wise Margins	`/reports`	Production Ready	Invoice-by-invoice margin percentage and profitability analysis
Ageing Analysis	`/reports`	Production Ready	Receivables and Payables aging buckets (0-30, 31-60, 61-90, 90+ days overdue)
Public Mini Web Shop	`/store/:companyId`	Production Ready	Customer-facing digital web catalogue with categories, cart, and "Order via WhatsApp"
Online Orders Pipeline	`/orders`	Production Ready	Fulfillment workflow: Pending → Accepted → Packed → Shipped → Delivered
WhatsApp Reminders	`/marketing`	Production Ready	Polite payment reminders with customer balance and direct dynamic UPI payment link
Promotions & Coupons	`/marketing`	Production Ready	Percentage and flat coupon codes with minimum order constraints and expiry rules
Data Backup & Restore	`/settings`	Production Ready	Complete 15-collection JSON snapshot export, file restore, and 1-click sample data seeder
Multi-Tenant RBAC	`/superadmin`, `/settings`	Production Ready	Super Admin dashboard, company partitioning, and 6 role-based permission tiers

3. Standalone Point of Sale (POS) Fast Billing

3.1 Terminal Hardware & Layout

- **Fullscreen Kiosk Mode:** Dedicated viewport terminal (100vw × 100vh) with no navigation sidebar to prevent cashier distraction.
- **Top Control Bar:** Live digital clock, store status badge, cashier initial avatar, held carts queue count, sound toggle, and terminal settings dialog.
- **Zero-Click Continuous Barcode Scanning:** Compatible with any standard USB, Bluetooth, or 2.4GHz handheld scanner operating in HID keyboard wedge mode. When a barcode is read:
- Matches product by Barcode → SKU → Product Name → Unique Match.
- Emits an audible confirmation tone via the Web Audio API.
- Increments cart quantity by +1 (checking against available stock limits).
- Immediately clears the search input for hands-free continuous item ringing.
- **Keyboard Hotkeys:**
- F2: Instant focus on search input.
- F4: Open Pay & Tender split-payment dialog.
- F7: Put current cart on hold.
- F9: Quick Cash 1-tap checkout.
- Esc: Close open modal windows.
- **Fast Cash Presets:** Instant denomination buttons (₹100, ₹200, ₹500, Exact Cash).
- **Split Tender Payments:** Customer bills can be split across Cash, UPI / QR, and Card / POS Machine on the same transaction.
- **Thermal Slip & Bill Generation:** Instant receipt modal and automated thermal printing formatted for 80mm (3-inch), 58mm (2-inch), or standard A4 invoice layouts.

---

4. Sales, Challans, Quotes & Returns

4.1 Tax Invoices & Place of Supply

- Configurable invoice sequence prefixes (e.g., INV-2026-001).
- Automatic tax detection: Applies CGST + SGST for Intra-State or IGST for Inter-State sales based on customer state vs. company state.
- Dynamic UPI QR code generated on invoice footer with payee VPA, amount, and invoice reference.

4.2 Estimates & Quotations Pipeline

- Create draft price quotations for client approval.
- One-click **Convert to Invoice** automatically generates the invoice and deducts physical inventory.

4.3 Delivery Challans & Dispatch Slips

- Issue dispatch vouchers with vehicle transport registration numbers, driver details, and delivery addresses.
- Goods remain tracked as "In-Transit" without creating an immediate debt journal entry until converted into a bill.

4.4 Credit & Debit Notes

- **Credit Notes (Sales Returns):** Reverse customer sales, re-stock inventory, and adjust customer accounts.
- **Debit Notes (Purchase Returns):** Reverse inward vendor purchases and deduct outstanding payable dues.

5. Inventory, Batches, Units & Warehouses

5.1 Product Master & Dual Units

- Supports base units (pcs, kg, ltr, box, pkt, m) and secondary units with defined conversion rates (e.g., 1 Box = 12 pcs).
- Tracks HSN/SAC codes, brand names, subcategories, buying cost, and selling price.
- Live stock balance with automated low stock threshold notifications.

5.2 Batch & Expiry Date Management (FEFO)

- Tracks Batch Number, Manufacturing Date, and Expiry Date per SKU.
- Visual alerts: Amber warning for stock expiring within 30 days; Red alert for expired stock.
- Enforces First-Expiry-First-Out (FEFO) picking guidelines.

5.3 Serial Number / IMEI Tracking

- Comma-separated unique serial number identification for electronic goods and serialized assets.

5.4 Stock Adjustments & Damage Audits

- Record inventory count revisions with categorized reason codes:
- *Damaged / Broken Goods*
- *Expired Goods*
- *Theft / Unaccounted Loss*
- *Physical Audit Variance*
- *Internal Store Consumption*

5.5 Multi-Godowns & Inter-Branch Logistics

- Define multiple warehouse or retail branch physical locations.
- **Stock Transfer Vouchers:** Log goods transfer from Source Godown to Destination Godown with transfer numbers.

5.6 Barcode & QR Label Designer

- Generate printable sticker sheets in Code128 barcode or QR format.
- Layout presets: Compact (40-Up), Standard (24-Up), and Large (12-Up) per sheet.

---

6. Accounting, Ledgers, Banking & GST

6.1 Cash Book & Multi-Bank Accounts

- Monitor petty cash registers, commercial bank current/savings accounts, and digital merchant wallets.
- **Contra Entries:** Account-to-account transfers (Cash Deposit, Cash Withdrawal, Bank-to-Bank).

6.2 Operating Expenses & ITC

- Log operational overheads (Rent, Utilities, Staff Salaries, Freight, Repairs).
- Toggle GST Input Tax Credit (ITC) eligibility with vendor GSTIN recording.

6.3 Party Ledgers & CRM

- Unified customer and supplier directories with GSTIN, billing addresses, credit limits, and credit periods.
- Interactive **Party Statement of Account** with live running Debit (Dr) / Credit (Cr) balances.

6.4 GST Compliance & E-Way Bills

- **GSTR-1:** Complete tables for B2B (Section 4A), B2C Large & Small (Section 7), CDNR (Section 9B), and Table 12 HSN Summary (quantities, taxable values, CGST, SGST, IGST breakdown) with direct NIC-ready JSON export.
- **GSTR-3B:** Consolidated Outward Liability vs. Eligible ITC summary.
- **NIC E-Way Bill:** Auto-formats consignment JSON for transport values > ₹50,000.
- **E-Invoicing:** Generates 64-character IRN hash codes and B2B signed QR code simulation.

---

7. Reports, Storefront & Administration

7.1 Business Reports & Financial Intelligence

- **Profit & Loss Statement:** Real-time Gross Sales Revenue (A), Direct COGS, Gross Profit, Operating & Administrative Overhead Expenses (B), and Net Profit Before Tax (A - B).
- **Balance Sheet:** Total Liquid Assets, Inventory Valuation, Receivables vs. Payables, and Owner Equity.
- **Bill-wise Profit Margins:** Margin percentage breakdown per invoice.
- **Ageing Analysis:** Accounts receivable and payable aging buckets (0-30, 31-60, 61-90, 90+ days).

7.2 Digital Web Shop & WhatsApp Orders

- Public web catalog (/store/:companyId) where customers browse products, view pictures, and submit orders.
- Orders management pipeline: Pending → Accepted → Packed → Shipped → Delivered.
- Automated WhatsApp payment reminders with balance summary and direct UPI payment links.

7.3 System Administration & Backup

- **1-Click Sample Business Data Seeder:** Instantly populate realistic demo products, invoices, purchases, bank accounts, godowns, and customers for testing.
- **Full Database JSON Backup & Restore:** Complete export and restore of all 15 Firestore collections.

  - **Role-Based Access Control (RBAC):** Granular permission isolation between Super Admin, Admin, and Staff/Cashier roles.